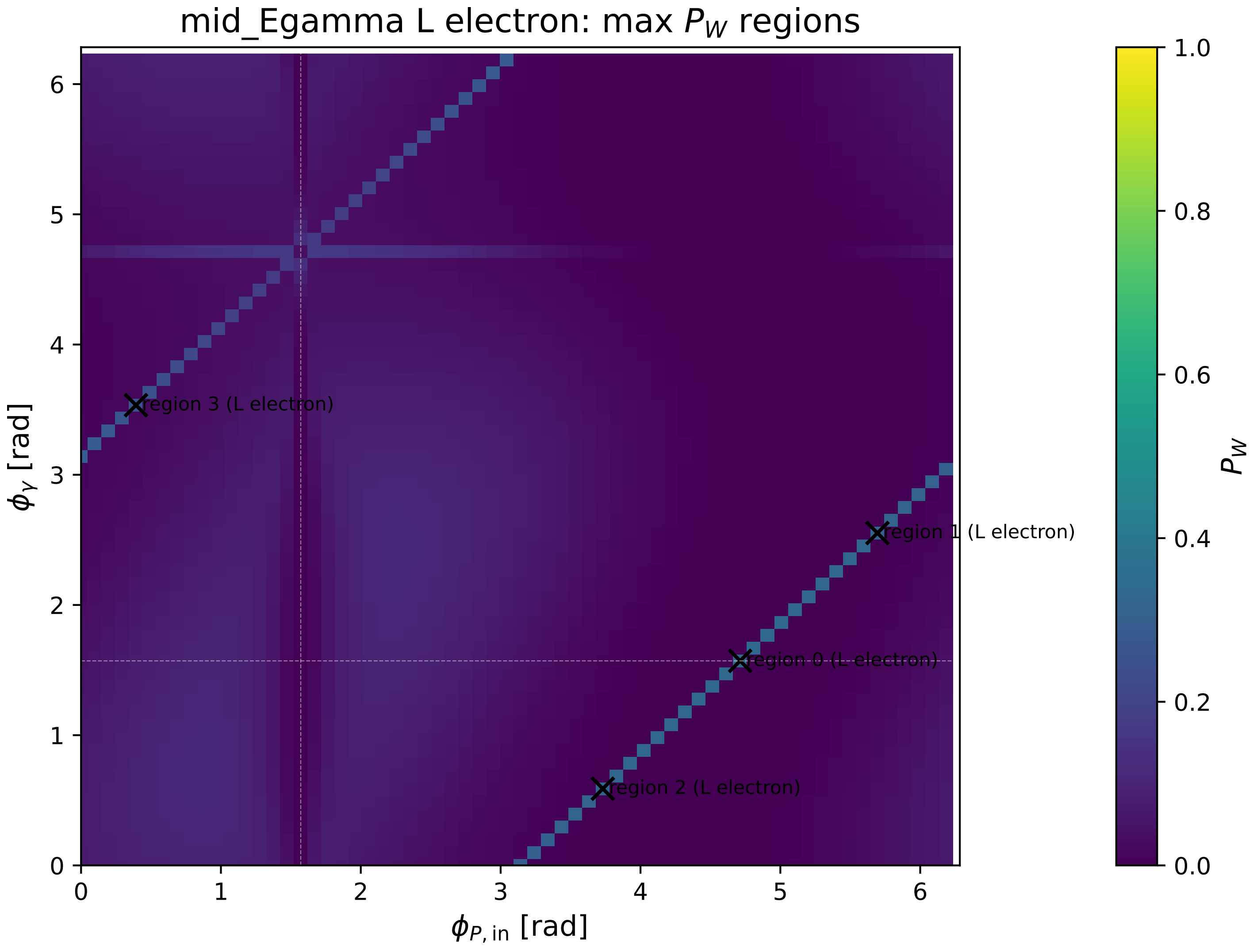
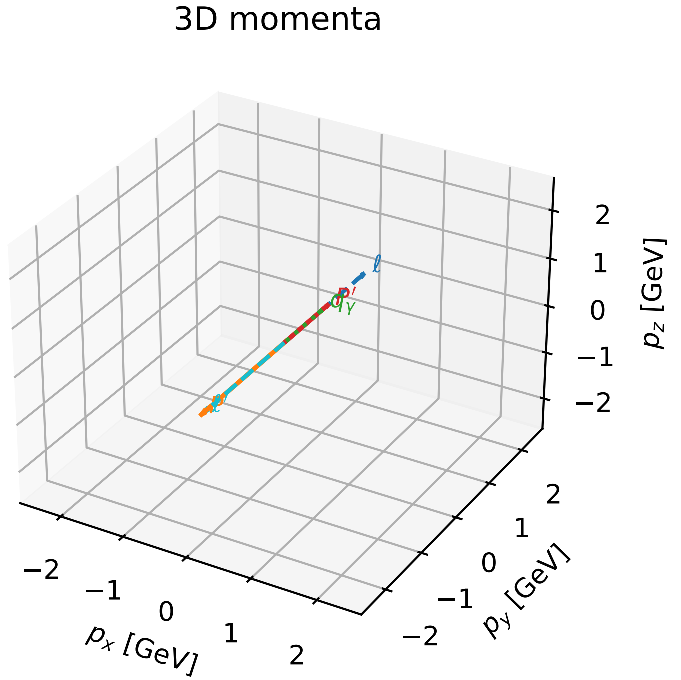


mid_Egamma L electron: max P_W regions
region 3 (L electron)
region 1 (L electron)
region 0 (L electron)
region 2 (L electron)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
P_W
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

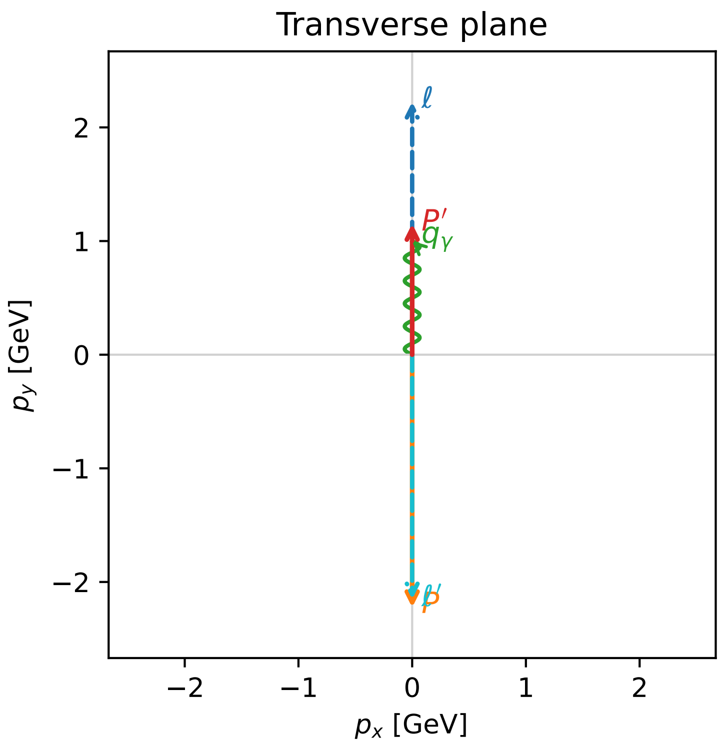
L electron characteristic max P_W configuration

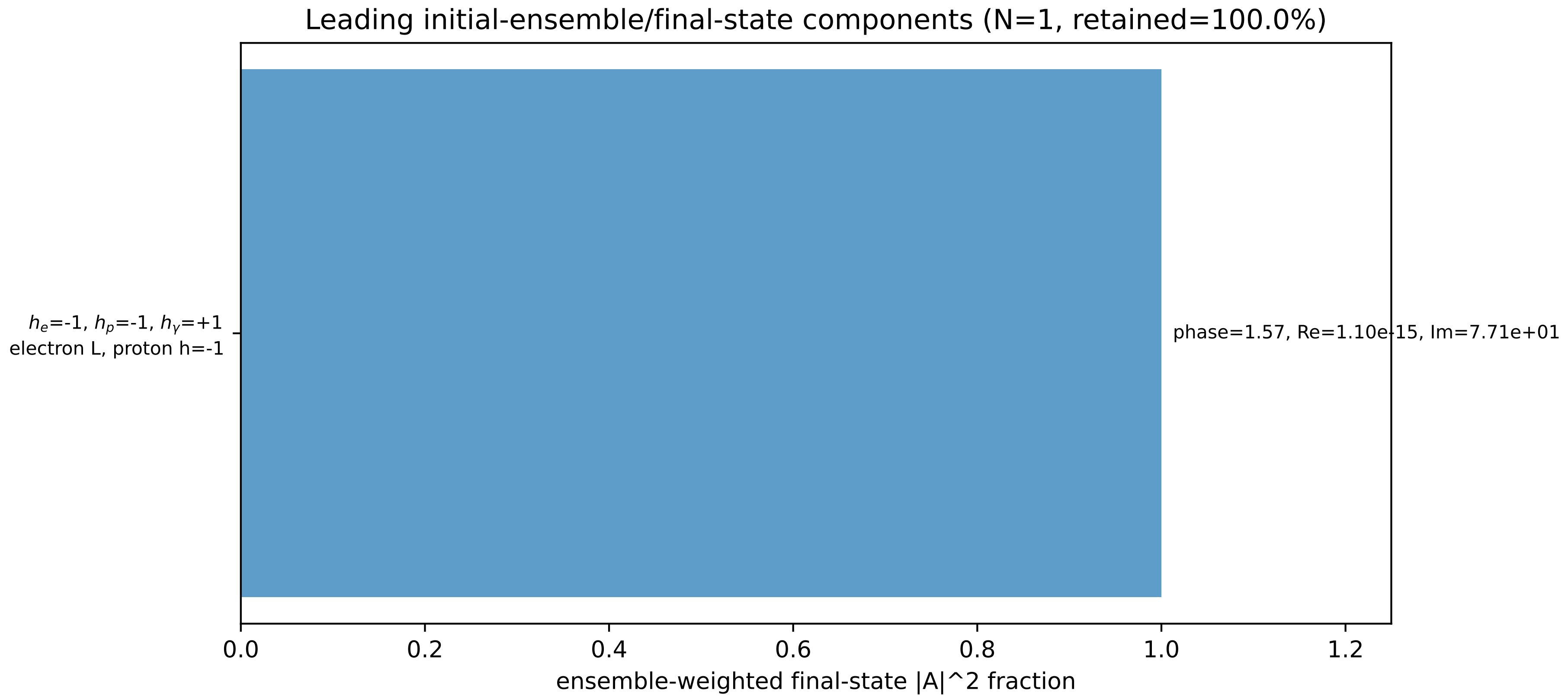


w_purity_mid_egamma_l_lepton_region_0 P_W=0.333262
region: mid_Egamma_L_lepton_region_0
outgoing-state purity: 0.99957
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.15253
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=1, ϕ_γ=1.5708
Q^2=19.1525, x_B=0.96635, t=-10.5424, W^2=1.54677, y=0.960429
$\theta(\ell', \gamma)$=180 deg

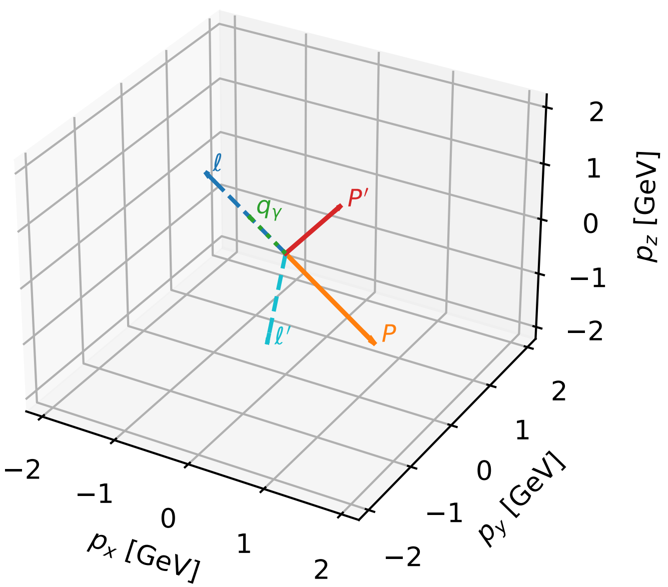
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.1525 p_x= -0.0000 p_y= -2.1525 p_z= 0.0000
P' E= 1.4860 p_x= 0.0000 p_y= 1.1525 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.0000 p_y= 1.0000 p_z= 0.0000





L electron characteristic max P_W configuration

3D momenta

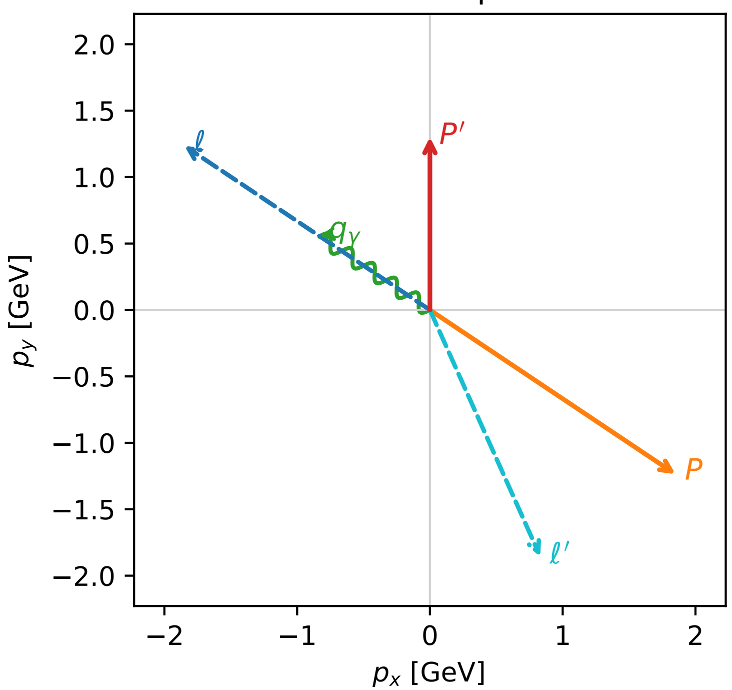


w_purity_mid_egamma_l_lepton_region_1 P_W=0.321672
region: mid_Egamma_L_lepton_region_1
outgoing-state purity: 0.975572
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

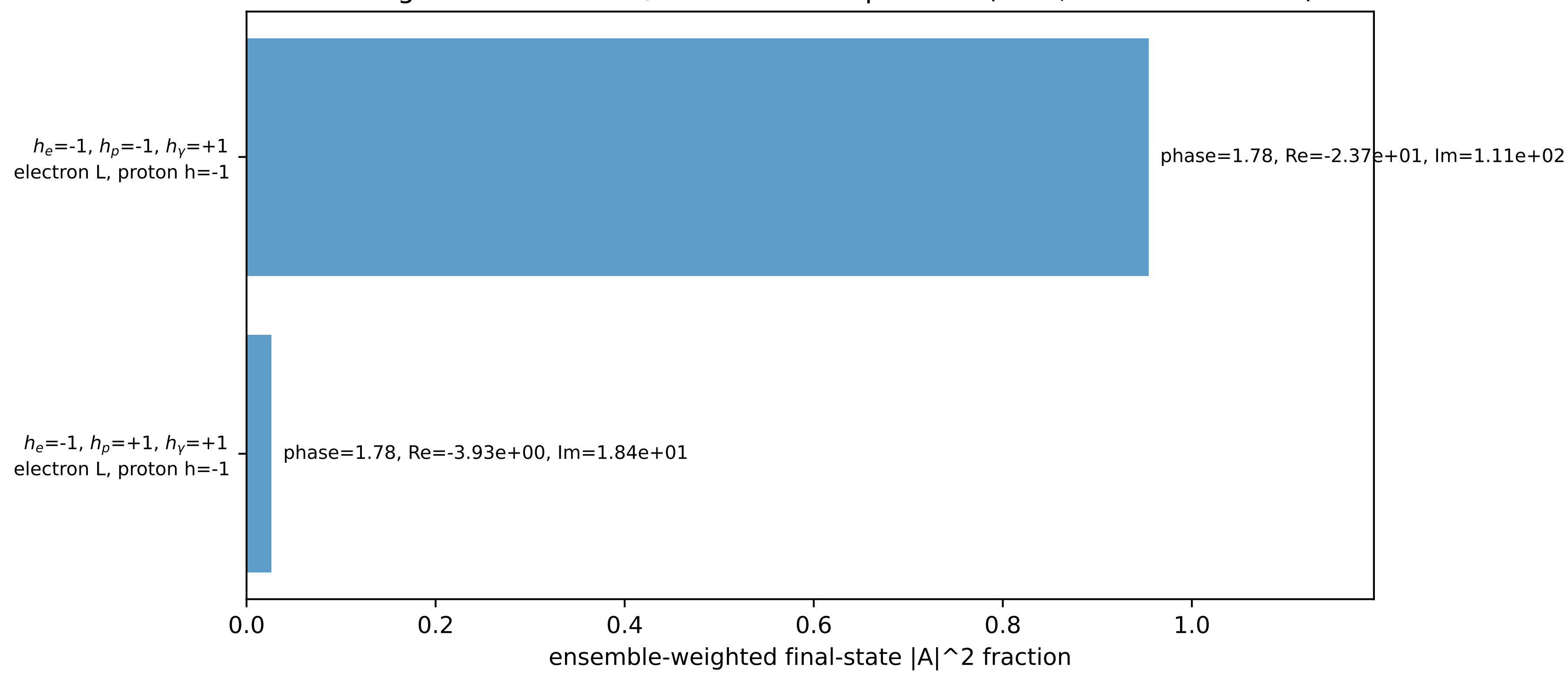
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.30122
θ_{in}=1.57079, ϕ_ℓ=2.55254, ϕ_P=5.69414
E_γ=1, ϕ_γ=2.55254
Q^2=16.7159, x_B=0.904629, t=-9.20119, W^2=2.64213, y=0.895436
$\theta(\ell', \gamma)$=147.873 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= 1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.0345 p_x= 0.8315 p_y= -1.8568 p_z= 0.0000
P' E= 1.6041 p_x= 0.0000 p_y= 1.3012 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.8315 p_y= 0.5556 p_z= 0.0000

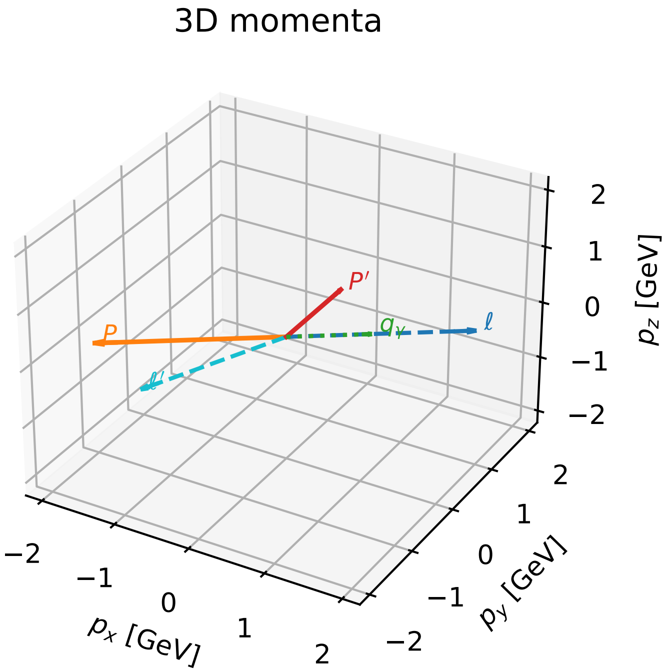
Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=98.0%)



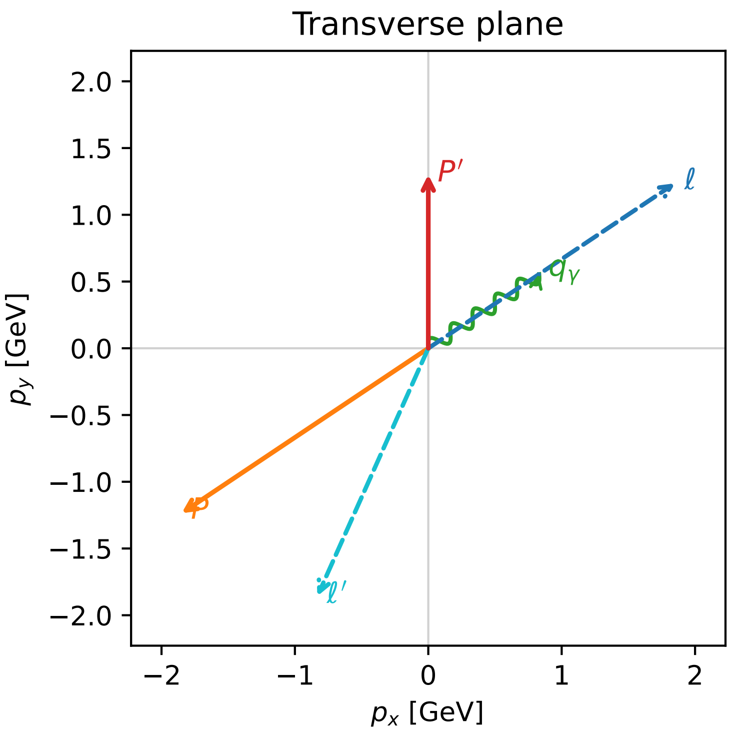
L electron characteristic max P_W configuration

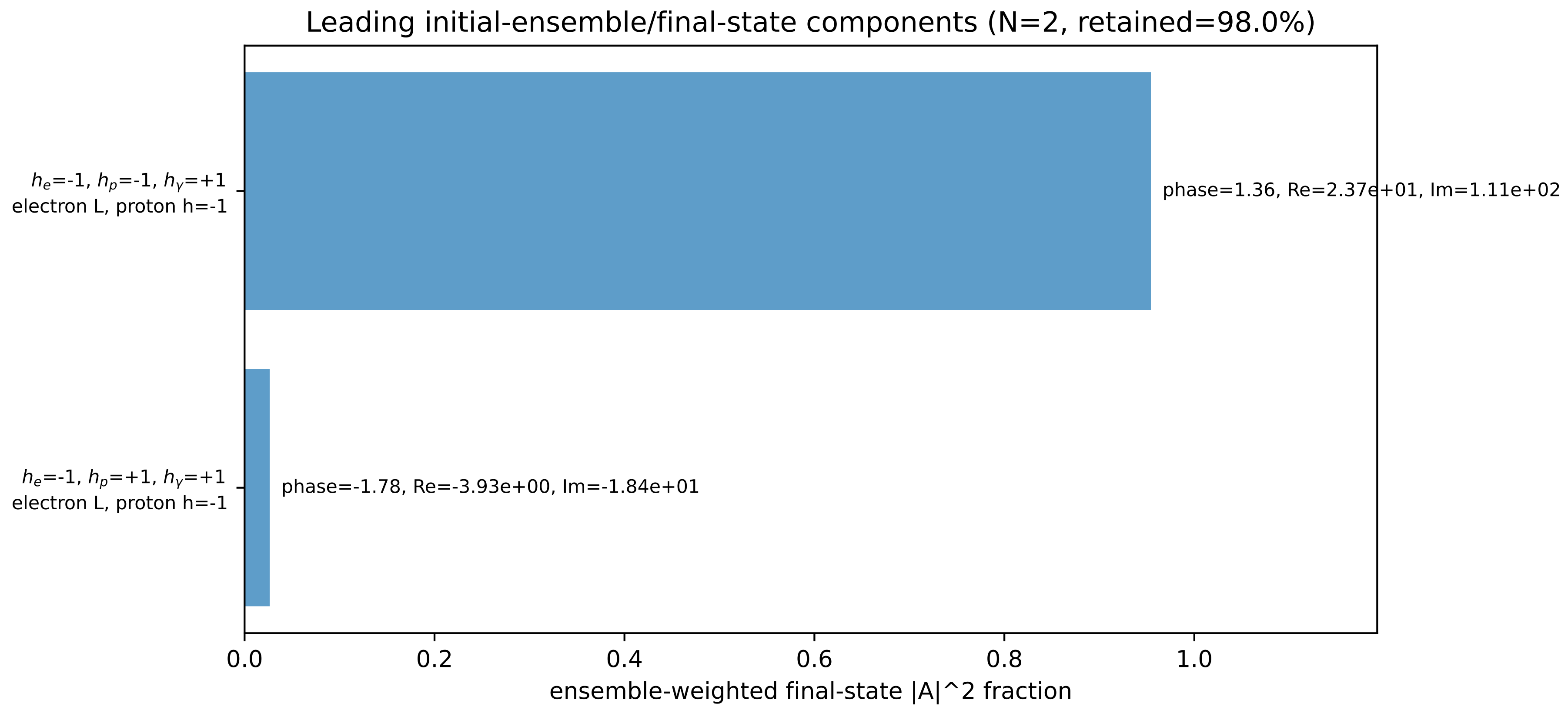


w_purity_mid_egamma_l_lepton_region_2 P_W=0.32124
region: mid_Egamma_L_lepton_region_2
outgoing-state purity: 0.975572
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

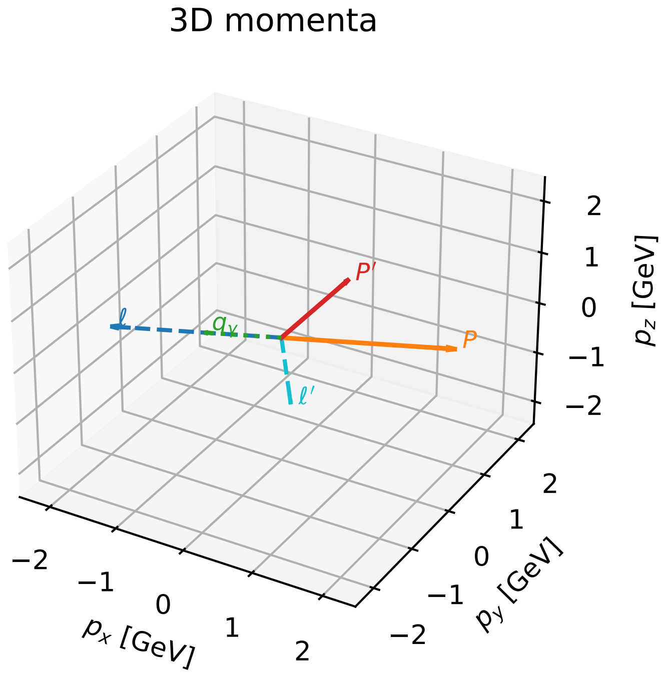
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.30122
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=1, ϕ_γ=0.589049
Q^2=16.7159, x_B=0.904629, t=-9.20119, W^2=2.64213, y=0.895436
$\theta(\ell', \gamma)$=147.873 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= -1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.0345 p_x= -0.8315 p_y= -1.8568 p_z= 0.0000
P' E= 1.6041 p_x= 0.0000 p_y= 1.3012 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.8315 p_y= 0.5556 p_z= 0.0000





L electron characteristic max P_W configuration

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w_purity_mid_egamma_l_lepton_region_3  P_W=0.258462
region: mid_Egamma_L_lepton_region_3
outgoing-state purity: 0.697639
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_e, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.75164
θ_{in}=1.57079, ϕ_ℓ=3.53429, ϕ_P=0.392699
E_γ=1, ϕ_γ=3.53429
Q^2=8.81411, x_B=0.623844, t=-4.85168, W^2=6.19444, y=0.684663
$\theta(\ell', \gamma)$=101.515 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= -2.0551	p_y= -0.8512	p_z= -0.0000
P	E= 2.4141	p_x= 2.0551	p_y= 0.8512	p_z= 0.0000
ℓ'	E= 1.6515	p_x= 0.9239	p_y= -1.3690	p_z= 0.0000
P'	E= 1.9870	p_x= 0.0000	p_y= 1.7516	p_z= 0.0000
q_γ	E= 1.0000	p_x= -0.9239	p_y= -0.3827	p_z= 0.0000